Lesson 6 — Transducers: Sensors and Actuators

ATmega328P / Arduino Uno | Source: lecture 06 slides, AH1815, ADXL335, BC547, 2N7000, 1N4001 datasheets, "Applications of Diodes" | Builds on Lessons 3–5

WHAT YOU WILL BE ABLE TO DO AFTER THIS LESSON

- Define a transducer, and tell sensors from actuators and digital sensors from analog ones.
- Explain how each sensor in the lecture works and how to connect it to the microcontroller.
- Explain why actuators cannot be driven straight from a pin, and drive them with a **BJT**, a **MOSFET** or a **relay**.
- Calculate a base resistor, and explain the flyback diode.

GIVEN TO YOU IN THE EXAM

Nothing on the three exam sheets is specific to this lesson apart from the pin functions (digital pins, A0–A5, PWM pins 3, 5, 6, 9, 10, 11 on the Arduino sheet). **Everything below is "learn it".** Datasheets of the actual parts are in `Lecture Materials/06 - ...`; you only need the few numbers quoted here.

1. What is a transducer?

A **transducer converts energy (or a physical quantity) from one form to another**. Transducers sit at the **boundary** of an embedded system: they are how it touches the real world. They are grouped by the **direction** of conversion:

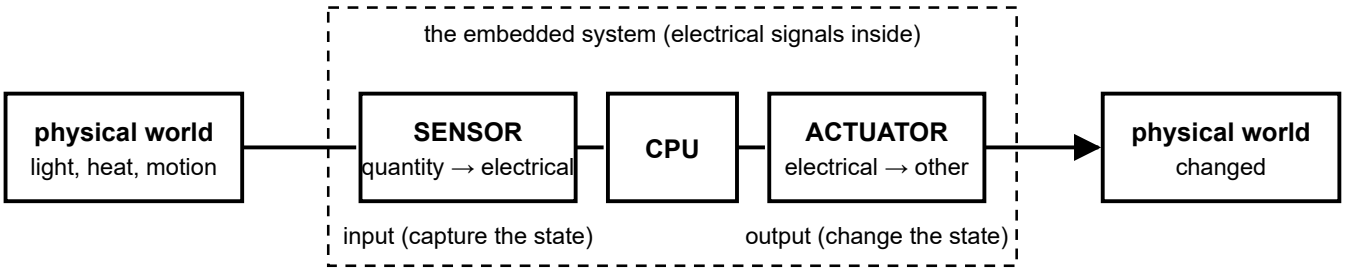


Figure 1. Sensors capture the state of the physical world as inputs; actuators change it.

SENSOR

Converts a physical quantity into an **electrical signal** (usually a voltage). Examples: LDR, LM35, reed switch, PIR, accelerometer.

ACTUATOR

Converts **electrical energy** into another form: light (LED, bulb), motion (motor, solenoid), heat/cold (Peltier), sound (buzzer).

2. Digital vs analog sensors

	Digital sensor	Analog sensor
Output	Two discrete voltages only: about 0 V or about 5 V ("yes/no": magnet present, motion seen)	A continuous voltage that follows the quantity (how bright, how hot, how tilted)
Connect to	Any digital I/O pin, read with `PINx` / `digitalRead`, or an interrupt pin	An ADC pin A0–A5, read with the ADC / `analogRead` (Lesson 5)
Lecture examples	Reed switch, Hall effect, PIR, flame	Water level, accelerometer, thumb joystick
Both	Sound, soil moisture, gas/smoke modules have an **Aout** (analog) pin **and** a **Dout** (digital) pin.	

How a module gives both Aout and Dout

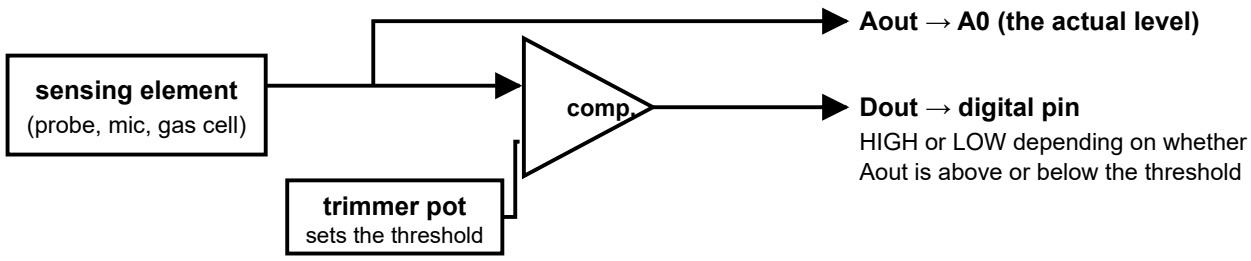


Figure 2. Inside an "analog/digital" module: a comparator (often an LM393 chip) compares the sensor voltage with a threshold you set with the small blue screw pot.

Use **Dout** when you only need "wet/dry", "loud/quiet", "gas/no gas" and want to set the threshold with a screwdriver. Use **Aout** when the program needs the level or must choose the threshold itself. Module pins are usually labelled **VCC (+5 V), GND, Aout, Dout**.

3. The digital sensors

Reed switch

Two thin metal contacts sealed in a glass tube with inert gas. **Normally open**; a **magnet** nearby magnetises the contacts so they attract and **close**. It is simply a switch operated by a magnet, so wire it exactly like a pushbutton: one side to the pin with a **pull-up**, the other side to GND.

Use: door/window alarms (magnet on the door, reed on the frame), position sensing, counting wheel turns.

Hall effect sensor (AH1815)

A semiconductor chip that produces a small voltage when a **magnetic field** passes through it (the Hall effect); built-in electronics turn this into a clean on/off output. The AH1815 is **omnipolar** (reacts to a north or south pole), runs on 2.5–5.5 V, and has an **open-drain output**: with a magnet the output is **pulled LOW**, otherwise it is "off" (floating). So it **needs a pull-up resistor** (internal pull-up or 10 kΩ to 5 V). No moving parts, so it is faster and lasts longer than a reed switch.

Pins (flat face towards you): VDD, GND, OUTPUT — check the datasheet drawing for your package.

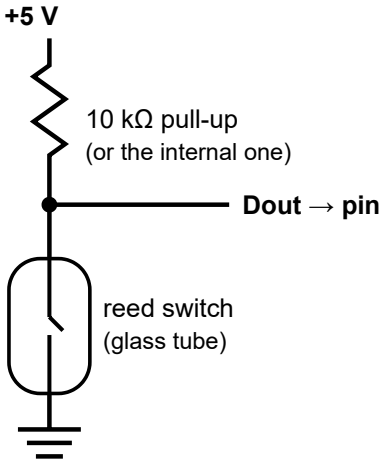


Figure 3. Reed switch with pull-up (same circuit for an open-drain Hall sensor: its output replaces the switch).

OPEN-DRAIN / OPEN-COLLECTOR, IN PLAIN WORDS

The output can only **connect the line to GND** (LOW) or **let go** (nothing). It cannot push the line HIGH by itself, so a pull-up resistor provides the HIGH. It lets the sensor work with any logic voltage.

PIR sensor (motion)

Pyroelectric **I**nfra**R**ed. Warm bodies give off infrared. A pyroelectric element produces a voltage when the IR falling on it **changes**; a white **Fresnel lens** splits the view into zones, so a person moving across zones causes changes. It detects **movement**, not presence: a person standing still is not seen. The module output goes **HIGH** (about 3.3 V, still read as HIGH) for a few seconds after motion. Modules have two trimmers (sensitivity and on-time) and need about **30–60 s to settle** after power-up. Pins: **+5 V, Dout, GND**.

Flame sensor

An IR photodiode/phototransistor tuned to the infrared emitted by a **fire**, with a comparator giving a digital output (often **LOW = flame**, check your module's LED). Use: fire alarms (the slide's example of a microcontroller system with binary inputs and outputs).

4. The analog sensors

Sensor	How it works	Connect / read
Water level	Exposed parallel copper traces. Water between them conducts; the more traces covered, the lower the resistance, so the output voltage rises with the water level.	Aout → A0. Power it only while reading (electrolysis corrodes the traces).
Accelerometer (ADXL335)	A tiny suspended mass inside the chip moves under acceleration; capacitive plates sense how far. Gives **three voltages**, one per axis X, Y, Z. At rest it measures **gravity**, so it senses **tilt**. Output is **ratiometric**: at 0 g each axis sits at half the supply (1.5 V at 3 V), about **300 mV per g**.	X, Y, Z → A0, A1, A2. The chip runs on 1.8–3.6 V; use a module with a regulator for 5 V (check yours).
Thumb joystick	Two potentiometers (X and Y) moved by the stick, plus a push switch when pressed down. Centre ≈ 2.5 V (ADC ≈ 512).	VRx → A0, VRy → A1, SW → digital pin with pull-up.
Sound	Microphone + amplifier: voltage follows the sound wave; Dout via comparator for "loud".	Aout → A0 or Dout → pin.
Soil moisture	Two probes in the soil: wet soil conducts better. Many modules give a **lower** Aout when wet; check by testing.	Aout → A0, or Dout.
Gas / smoke (MQ series)	A heated sensing layer whose resistance falls in the presence of gas. The heater needs a warm-up time and draws noticeable current.	Aout → A0, or Dout.
LDR, LM35, pot	Covered in Lesson 5 (voltage divider, 10 mV/°C).	A0–A5.

Accelerometer example: board flat: Z feels +1 g, X and Y 0 g. Tilt so X points down: X reads +1 g. With 3 V supply: X voltage = 1.5 V + 0.3 V × g. ADC on 5 V reference: 0 g ≈ 1.5 × 1024 / 5 = 307, 1 g ≈ 1.8 V ≈ 369.

5. Actuators, and why a pin cannot drive them directly

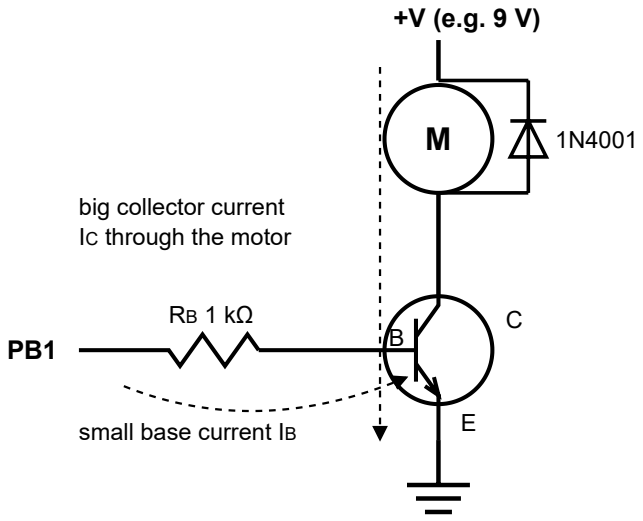
An I/O pin can supply only **5 V** and about **20 mA** (40 mA absolute maximum, and the whole chip about 200 mA). An LED (15 mA) is fine. But:

Actuator	Converts to	Needs
RGB LED	red, green, blue light; mixing gives other colours	Fine from pins: three LEDs in one package, one resistor and one pin each (PWM pins to mix shades, Lesson 8). **Common cathode**: shared leg to GND, pin HIGH = on. **Common anode**: shared leg to 5 V, pin LOW = on.
Incandescent bulb	light and heat	Higher voltage and current: transistor or relay.
DC motor	circular motion	Even small ones draw 100–500 mA, and more when starting: transistor/MOSFET + flyback diode, often its own supply.
Solenoid	straight-line motion (a plunger)	Higher voltage and current: MOSFET + flyback diode.
Thermoelectric (Peltier)	heat on one face, cold on the other	Amps of current: MOSFET or relay, separate supply.

THE RULE

The pin only **controls** a switch; a separate, stronger supply **powers** the load. The switch is a **BJT** transistor, a **MOSFET**, or a **relay**. The grounds of the two supplies must be joined, or the control signal has no reference.

6. Driving a load with a BJT (BC547)



GND of the 9 V supply joined to the Uno GND

Figure 4. Low-side switch: the load sits between +V and the collector; the emitter goes to GND.

How a BJT works: it has three legs, Base, Collector, Emitter. A **small current into the base** allows a **much bigger current from collector to emitter**. The ratio is the current gain $\mathbf{h_{FE}}$ (110–800 for BC547): $I_C = h_{FE} \times I_B$. As a switch we give **more** base current than needed, so the transistor is fully on (**saturated**: only about 0.2 V across C–E) or fully off (no base current).

Water picture: the base is a small tap controlling a big valve.

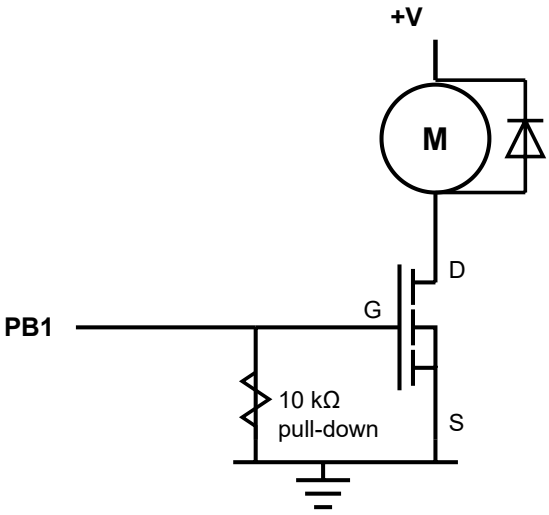
BASE RESISTOR CALCULATION

1. Load current I_C, e.g. relay coil 70 mA. 2. Minimum base current = $I_C / h_{FE(min)}$ = 70 / 110 = 0.64 mA. 3. Multiply by about 5 for hard saturation: $I_B \approx 3.2$ mA. 4. $R_B = (V_{pin} - V_{BE}) / I_B = (5 - 0.7) / 0.0032 = 1.3\ k\Omega \rightarrow$ use **1 kΩ** (gives 4.3 mA, still safe for the pin).

BC547 limits: I_C max **100 mA**, V_{CE} max 45 V. A bigger motor needs a bigger transistor or a MOSFET. **Pins (flat face towards you):** C, B, E.

Why "low side"? With an NPN the emitter must be at GND so the 5 V pin can push current into the base. The load goes on the "high" (+V) side.

7. Driving a load with a MOSFET (2N7000)



gate takes almost no current, so no gate resistor is needed for

Figure 5. N-MOSFET as a low-side switch.

How a MOSFET works: three legs, Gate, Drain, Source. It is controlled by **voltage**, not current: when the gate is more than the **threshold voltage** (0.8–3 V for 2N7000) above the source, a channel opens and current flows from drain to source. The gate is insulated, so it draws **almost no current** (only a tiny charge when switching).

The 10 kΩ pull-down keeps the gate at 0 V (MOSFET off) while the microcontroller is resetting and its pins are still floating inputs. Otherwise the motor might twitch at power-up.

2N7000 limits: about 200 mA continuous, 60 V. For motors of several amps use a "logic-level" power MOSFET (one fully on at V_{GS} = 5 V). **P-MOSFET**: the opposite type, used on the high (+V) side; gate pulled LOW to turn on.

	BJT (BC547)	MOSFET (2N7000)
Controlled by	base **current**	gate **voltage**
Needs	base resistor	pull-down (recommended)
Drop when on	~0.2 V ($V_{CE(sat)}$)	I × $R_{DS(on)}$ (a few Ω)
Best for	small loads, cheap	bigger loads, PWM, efficiency

8. The flyback diode: why every motor and relay needs one

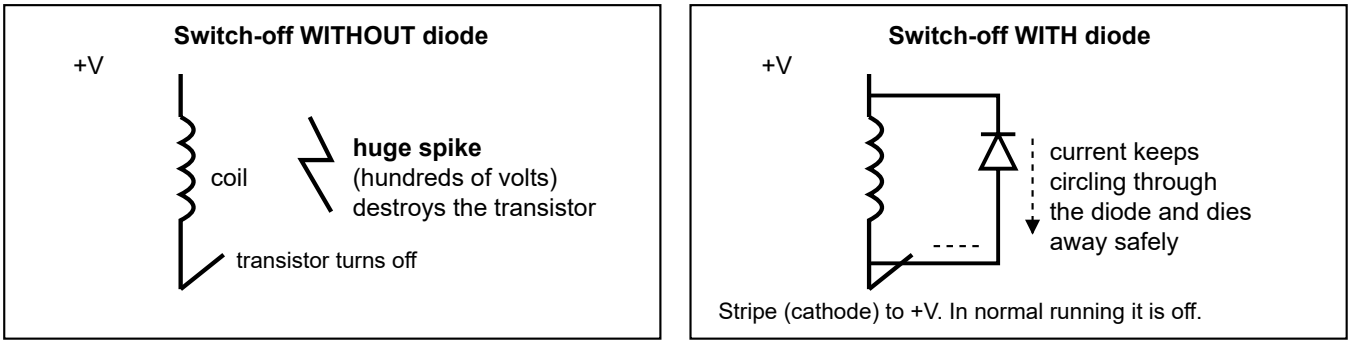


Figure 6. A coil (motor, relay, solenoid) stores energy in its magnetic field and resists any sudden change of current.

Why it happens: current through a coil cannot stop instantly. When the transistor suddenly opens, the coil forces the current to keep going by producing a very high voltage across the transistor. The **flyback (freewheeling) diode** placed across the coil, **cathode to +V**, gives that current a loop to circulate in until it dies out. It is "reverse-biased" (off) while the load runs normally. The 1N4001 (1 A, 50 V, ~0.7–1.1 V drop) is the standard choice. **Other diode uses** (web resource): rectifying AC to DC, protecting a circuit from a reversed battery, and LEDs themselves.

9. Switching with a relay

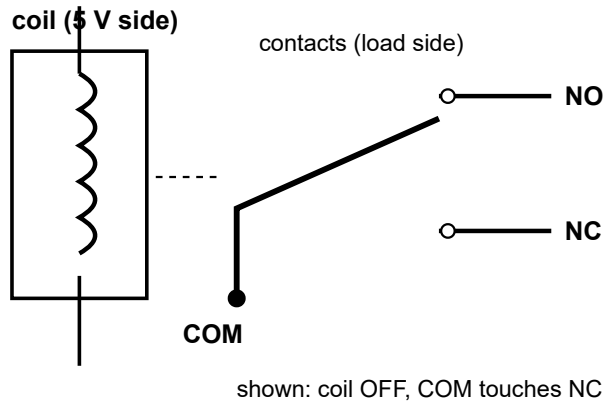


Figure 7. A relay: an electromagnet that moves a switch. COM = common, NO = normally open, NC = normally closed.

How it works: current through the **coil** makes a magnetic field that pulls the moving contact from **NC** over to **NO**. Coil off: a spring pulls it back. You hear a click.

Why use one: the coil circuit and the contact circuit are **electrically isolated**, so 5 V logic can switch **230 V AC** or large DC loads. The contact rating (e.g. 10 A 250 VAC) is printed on the case.

Driving it: the coil needs more current than a pin gives, so use the BJT circuit of section 6 with the coil as the load and a flyback diode across it. A **relay module** already contains the transistor, flyback diode, an indicator LED and screw terminals: you connect VCC, GND and IN (to the pin).

Wiring a lamp: live (through a fuse) → COM, NO → lamp, lamp → neutral. Lamp lights only while the coil is on. Use NC instead for "on unless activated". **Never build mains circuits on a breadboard.**

Limits: slow (milliseconds), so not for PWM; contacts wear out.

10. Worked design questions (exam style)

A. Door alarm with a reed switch

Spec: when the door opens, an LED (or buzzer) turns on and stays on until reset. **Design:** magnet on the door, reed switch on the frame, wired from **PD2 (pin 2)** to GND with the internal pull-up. Door closed: magnet near, reed closed, pin LOW. Door open: reed opens, pin goes HIGH. LED on **PB0 (pin 8)** through 220 Ω.

```
void setup()
{
  pinMode(2, INPUT_PULLUP);       // reed switch
  pinMode(8, OUTPUT);             // alarm LED / buzzer
}
void loop()
{
  if (digitalRead(2) == HIGH)     // door opened
    digitalWrite(8, HIGH);        // latch on
}                                 // reset button clears it
```

```
#include <avr/io.h>
int main(void)
{
  DDRB  |=  (1<<PB0);
  DDRD  &= ~(1<<PD2);
  PORTD |=  (1<<PD2);             // pull-up
  while (1)
    if (PIND & (1<<PD2))          // HIGH = open
      PORTB |= (1<<PB0);          // alarm on
}
```

B. Fan that switches on when it is hot

Spec: LM35 on A0; a 6–9 V DC motor turns on at 30 °C and off at 28 °C. **Design:** motor powered from the 9 V battery through a 2N7000 (or BC547 if the motor draws under 100 mA) on **PB1 (pin 9)**, flyback diode across the motor, 10 kΩ gate pull-down, grounds joined. Two thresholds (hysteresis, Lesson 5) stop it chattering at 29 °C. Pin 9 is also a PWM pin, so Lesson 8 can later control the **speed** (Exercise 6).

```
void setup() { pinMode(9, OUTPUT); }
void loop()
{
  float t = analogRead(A0) * 5.0 / 1024 * 100;   // LM35: degrees C
  if (t >= 30)      digitalWrite(9, HIGH);       // hot: fan on
  else if (t <= 28) digitalWrite(9, LOW);        // cool: fan off
  delay(500);
}
```

11. Build example: a motor driven by a BC547

This is component test 09 in the `Component Tests` folder (wiring and pictures there). Use the Lesson 3 section P method. What is new for actuators:

1. **Two supplies:** the motor runs from the 9 V battery, never from the Uno 5 V pin. Join the battery **−** to the Uno GND rail. Never connect the battery **+** to the Uno 5 V or the breadboard + rail used by the Uno.
2. **Order of wiring:** transistor first (check C, B, E), then 1 kΩ from the pin to the base, emitter to GND, motor between battery + and collector, diode across the motor (stripe to battery +). Connect the battery **last**, after checking.
3. **Test in steps:** blink sketch on the pin with an LED in place of the motor first (proves the transistor), then the motor.
4. **Signs of a fault:** Uno resets when the motor starts (motor on the Uno supply); motor always on (C and E swapped or transistor damaged); transistor hot (diode reversed, or motor current above 100 mA).

L. What the lecturer explained in the lectures (2020 recordings)

From the two transducer recordings (folders 18–19 in `Embedded Systems Lectures`). **[EXAM]** marks things he tied to the exam.

L1. THE PRACTICAL EXAM, IN HIS WORDS [EXAM]

50% of the module is the practical exam and 50% the written paper. In the practical you get **3 hours** to **assemble the hardware, write the firmware and show a working system**. "If you talk to seniors, they will say the practical exam is the most difficult part of this module", because students don't practise. You can't do it in 3 hours without prior hands-on practice.

L2. Transducers: the idea

- A transducer converts energy from one form to another, in either direction. **Inward** (other energy → electrical) = **sensor**: LM35 (thermal), LDR (light), potentiometer (rotation). **Outward** (electrical → other) = **actuator**: LED (light), motor (motion), speaker or buzzer (sound).
- Transducers sit at the **boundary** of the embedded system.
- A digital sensor's two voltages are not exactly 0 and 5 V: roughly **0–0.8 V reads LOW, 2–5 V reads HIGH**. The pushbutton you already used is a digital sensor.
- If you understand these common sensors and how to connect them, you can connect any other sensor the same way.

L3. Digital sensors: points from class

Sensor	What he added
Reed switch (~Rs 25)	Contacts sealed in a glass tube filled with inert gas. Because any **spark stays inside the sealed tube**, reed switches are used in **dusty and hazardous places with flammable gases or liquids**, where an ordinary switch's spark could start a fire (his example: switching on a kitchen light after an overnight LP-gas leak). Also used in **door/window security**: reed on the frame, magnet on the door. Door closed → switch on → Dout LOW; opened → Dout HIGH, which signals a break-in. Read it with `PINx` in C or `digitalRead` in Arduino. With the internal pull-up, no external 10 kΩ is needed.
Hall effect	Same behaviour as a reed switch (Dout normally HIGH with a pull-up, LOW near a magnet) but **smaller, solid state, no mechanical contacts**, so it switches **faster**, has **no wear**, and lasts longer. Main use: **wheel-speed sensing in ABS**. A small magnet on the wheel hub passes the sensor once per turn, producing pulses; more pulses per second means a faster wheel, so you can work out the RPM. Stray magnetic fields don't matter, because the magnet must come very close.
PIR (~Rs 200)	**PIR = Pyroelectric InfraRed, not "passive infrared"**. He said 90 out of 100 people get this wrong. Output is **normally LOW** and goes **HIGH** when the IR energy in its field of view changes (a warm body moving). Pets and hot vehicle engines also trigger it; turn the sensitivity down to ignore cats and dogs. **It only detects movement *across* its field of view**: someone walking straight towards it is not seen, so mount it where people will cross in front of it. It can be fooled by hiding body heat (a thermal blanket or aluminium foil). It is used to switch on CCTV recording and floodlights.
Flame (~Rs 200)	Detects the IR a fire gives off (wavelength about **760–1100 nm**, different from body heat). Dout **normally HIGH, LOW when a flame is seen**; sensitivity set with the screw pot. The module's **LM393** chip is a comparator (an op-amp circuit) that turns the sensor's analog signal into a digital output. Used in fire alarms.

L4. Analog and analog/digital sensors: points from class

Sensor	What he added
Water level (~Rs 80)	Interleaved copper traces. More water bridges more traces, so resistance drops and Aout rises. Needs the ADC.
Accelerometer	Aout X, Y, Z are each proportional to acceleration along that axis. Used for **phone orientation**, **camera shake detection** and **drone motion**. The breakout board is for prototyping; phones use a tiny SMD chip.
Thumb joystick	Simply **two independent potentiometers**, one per axis, giving Aout X and Aout Y.
Sound (~Rs 150)	Aout is proportional to the sound's **amplitude**. Dout is **normally HIGH, LOW above the threshold** set by the pot. Used for **clap-operated** or **door-knock-operated** systems.
Soil moisture	Two fork-shaped probes; water bridges them, lowering the resistance. Dout is **normally HIGH, LOW when moisture is above the threshold**.
Gas / smoke	Pick the right model: **MQ2 = LPG** (~Rs 300), **MQ3 = alcohol**, **MQ4 = methane**; others detect carbon monoxide. Aout is proportional to concentration. Dout is **normally LOW, HIGH above the threshold**.

Watch the Dout direction, it differs by module: reed, Hall, flame, sound and soil are normally HIGH, going LOW on detection; PIR and gas are normally LOW, going HIGH on detection.

L5. Actuators: the numbers he used

- An ATmega328P pin gives 5 V and **at most 40 mA**. If a load tries to draw more, the pin heats up and burns out: "the motor doesn't care what the pin can deliver".
- **Kit motor (6–9 V)**: about **75 mA with no load**, more with a fan or wheel attached. Too much for a pin.
- **Solenoid**: straight-line motion (a plunger), about **12 V and 500–600 mA**. Typical use: an electric (even IoT) **door lock**.
- **Thermoelectric / Peltier** (named after Jean Charles Peltier): current moves heat from one face to the other, with about **70 °C** between the faces. Keep the hot side at 70 °C with a heat sink and fan, and the cold side reaches about 0 °C. A 2×2-inch module needs **12 V 5 A = 60 W** (~Rs 1500).
  Pros: no moving parts, no compressed gas, long life, no maintenance, one device for both heating and cooling.
  Cons: high cost per unit of cooling, and poor power efficiency compared with a compressor fridge.
  Uses: cooling overclocked CPUs and the IR sensor of heat-seeking missiles.
- **Incandescent bulb**: the filament gives light *and* heat. 12 V or 230 V and up to amps, so it cannot go on a pin.
- **RGB LED**: the **longest leg is the common cathode**. With it on the left, the others are R, G, B. Treat it like three LEDs, each with its own resistor. Mixing the three gives almost any colour.

L6. The BJT as he taught it

- **Bipolar** junction transistor: it has **two** P–N junctions. It is **current-driven**: a tiny base–emitter current (μA) lets a much larger collector–emitter current (mA) flow. In his demo the collector current was about 1000× the base current. More base current gives more collector current; zero base current turns it off. So it is an electronic switch, a small current controlling a large one (his water-gate analogy).
- **NPN** = P sandwiched between two N layers. You push current base → emitter, and the large current flows collector → emitter. **PNP** = N between two P layers; current flows emitter → base and emitter → collector. The **arrow on the emitter shows the current direction**.
- **BC547** datasheet: **pin 1 C, pin 2 B, pin 3 E**; collector current max **100 mA**; up to **45 V** between collector and emitter. Very cheap. Big **power transistors** handle amps and have a hole for bolting to an aluminium **heat sink**. Small ones need no heat sink.
- **Two separate circuits:** the pin supplies only the tiny base current (through the **1 kΩ**); the motor's current flows in the collector–emitter circuit from the **9 V battery**, whose − terminal is joined to the Uno GND (**common ground**). The pin never carries the motor current.
- **How to choose a transistor:** check two datasheet numbers against your load, the continuous collector (or drain) current and the maximum voltage. The kit motor (75 mA, 9 V) is fine on a BC547 (100 mA, 45 V). Don't buy a far bigger transistor than needed, as it just costs more.

L7. Flyback diode, in his words

When the transistor turns off, the motor keeps spinning for a moment from **momentum**. A coil spinning in a magnetic field acts as a **dynamo**, and the collapsing field (**self-induction**) produces a high-voltage spike in the opposite direction. Through the transistor, that spike destroys it. A diode across the motor, **anode to the collector side, cathode to +V**, gives the spike a safe path to "fly back". Connect it the other way and there is no protection. It must handle the current and voltage: use a **1N4001 (1 A, 50 V)**, not a small signal diode. Same effect as the **choke in a fluorescent tube light**, which makes the high voltage that starts the tube when the starter opens. Reading: `Web Resources/06 - Applications of Diodes.md`.

L8. MOSFET as he taught it

- **M**etal **O**xide **S**emiconductor **F**ield **E**ffect **T**ransistor. Pins: **Gate, Drain, Source**. **Voltage-driven**: the gate takes a voltage, not a current, "pressure, not flow" in his water analogy.

- **Why modern designs prefer MOSFETs:** much **faster switching** and **less heat** (important in small wearable and portable devices). Downside: **more expensive** (2N7000 about Rs 10–15 vs BC547 a few rupees).
- **N-channel:** a positive gate voltage lets current flow drain → source. **P-channel:** a negative gate voltage, current flows source → drain.
- **No gate resistor** is needed in his circuit: no current flows into the gate, so there is nothing to limit.
- **2N7000**: about **200 mA** drain current and **60 V**, plenty for the 75 mA / 9 V motor. Still needs the flyback diode.
- BJTs and MOSFETs **look identical** (same packages): you can only tell them apart by the part number and datasheet.

L9. Relays, as he taught them

- A relay is a **spring-loaded electromagnetic switch**. Current in the coil makes a magnetic field that pulls the contact over. When the current stops, the field collapses and the spring pulls the contact back. Terminals: **COM** (common), **NC** (normally closed), **NO** (normally open, closes when the coil is energised).
- Used for **high voltage and current**, e.g. a 230 V AC conveyor motor drawing 5–10 A, switched by a 5 V coil.
- Even a small 5 V relay coil needs **about 70 mA**, more than a pin's 40 mA. So drive it with a **BC547 + 1 kΩ + 1N4001 flyback diode** (the coil is an inductor too).
- A **relay breakout board** (~Rs 100–125) already has the relay, transistor, resistor, flyback diode, indicator LED and screw terminals. Still know how the circuit works, in case you have to build it.
- **Rule of thumb:** use a **relay for 230 V AC** loads, a **transistor/MOSFET for DC** loads.
- **Safety:** avoid experimenting with mains (230 V can be fatal) unless you are trained. Practise relays with DC supplies (a battery or wall adapter) instead.
- **Ideas he gave:** an **auto light** (LDR + relay switching a 230 V lamp at nightfall) and an **auto fan** (LM35: fan on above 30 °C, off below).

12. Common mistakes

Mistake	Result / fix
Motor or relay straight on a pin	Pin overloaded, chip may be damaged. Always a transistor/MOSFET.
No flyback diode on a coil	Spike kills the transistor, resets the chip. Diode across the coil, stripe to +V.
Grounds not joined	The transistor never switches. Join the load supply GND to the Uno GND.
No base resistor	Huge base current: pin and transistor damaged.
No pull-up on reed / open-drain Hall	Input floats, random readings.
Expecting PIR to see a still person	It detects movement only; also wait for warm-up.
3.3 V-only sensor on 5 V	Can be damaged (bare ADXL335). Check the module voltage.
Load current above the part's limit	BC547 > 100 mA or 2N7000 > 200 mA overheats. Pick a bigger part.

Cheat sheet

Sensors	
Digital	2 levels, digital pin. Reed, Hall, PIR, flame
Analog	continuous, A0–A5. Water level, accel., joystick
Both	Aout + Dout (comparator + trimmer). Sound, soil, gas
Reed / Hall	magnet → pin LOW (with pull-up)
PIR	movement → HIGH a few s; warm-up 30–60 s
ADXL335	0 g = Vs/2, ~300 mV/g, 3 axes

Actuator drivers	
Pin limit	5 V, ~20 mA (40 mA max)
BJT	$I_C = h_{FE}\, I_B$; $R_B = (5 - 0.7) / I_B$; BC547 ≤ 100 mA
MOSFET	gate voltage > threshold; 10 k pull-down; 2N7000 ≤ 200 mA
Flyback	diode across coil, stripe to +V
Relay	coil vs contacts isolated; COM / NO / NC
Always	separate load supply, grounds joined